

Assignment 4

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1. Introduction

In this assignment, we developed a character-level Transformer language model, trained it on a corpus, and explored its attention mechanism. The goal was to understand how the model processes and attends to different parts of input sequences through its multi-head attention layers. We focused particularly on interpretability and examined attention patterns across various layers to identify which linguistic structures the model focuses on.

2. Model Development

We used a Transformer architecture with the following specifications:

- Embedding size: 192
- Number of layers: 6
- Number of heads: 6
- MLP hidden size: 768
- Dropout: 0.1

The model was trained on character-level sequences using the CharTokenizer.

3. Training Procedure

The model was trained on both **English** and **Hebrew** datasets with the following hyperparameters:

- Learning rate: 5e-4
- Weight decay – 1e-3
- Batch size: 128
- Temperature: 0.7
- Top k sampling: K=5
- Optimizer: AdamW

The training loss results for both English and Hebrew data are shown below:

Loss Results (English and Hebrew Data):

Dataset	Loss
English	0.72
Hebrew	0.35

Samples:

```
[49800] Loss: 0.7446
Sample: w the our tenting thousand steel
Turn the trade the seas of death in such general.

HORTENSIO:
'Tis well; and must he die, sir:
He most is a game of fearful head; there's
not am I at thou art else to

[49810] Loss: 0.7518
[49820] Loss: 0.7379
[49830] Loss: 0.7446
[49840] Loss: 0.7418
[49850] Loss: 0.7176
[49860] Loss: 0.7314
[49870] Loss: 0.7291
[49880] Loss: 0.7187
[49890] Loss: 0.7463
[49900] Loss: 0.7305
Sample: w you have stay'd here are wretched for the war,
And his three hundred chostly crosses,
Whose haughty six wrath winten, I mean this sigh
To be my poor free unsuspected and
And said 'We were i' the sun

[49910] Loss: 0.7182
[49920] Loss: 0.7305
[49930] Loss: 0.7232
[49940] Loss: 0.7296
[49950] Loss: 0.7290
[49960] Loss: 0.7443
[49970] Loss: 0.7378
[49980] Loss: 0.7472
[49990] Loss: 0.7399
[50000] Loss: 0.7244
Sample: w thee, my heart for a world more.

PRINCE EDWARD:
That's my father's door, I fear thy too.

KING HENRY VI:
Threaten pitchers have my like my father's land,
To make a power there, to such a child,
Thi
```

נפוצץ ושאר,

אך יש בקרב קצרה ידי עולה נראה לי:

"היש לפו-טו-ריזם.

מגע אל שעורה"...?

.

רחל שתקה נכלמת,

ותאנח מר.

והנה על מאלמת.

שיחה נסבה כבר...

לשירת רחל.

לדף הראשי של פרויקט בן-יהודה.

ד

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ברכת היום.

הקין.

הקין.

מחין.

יש מעיר ויר.

ויאכר שני.

חור, קודר הושע-נא ואבל.

לחיות רחוקה מגבוליד. –

תתניני, כנרת,

לנוח בבית-קברותיד.

אין דמי-חדלון.

בעצב שלו הרוגע,

ושיר-עבוד

4. Attention Mechanism Analysis

We conducted a detailed attention mechanism analysis to explore what the model learns at each layer, particularly focusing on Layer 1's attention heads.

We generated heatmaps to visualize attention weights across the attention heads.

We observed the following patterns across the heads when analyzing **Layer 1** for the prompt 'If a then b. If b then c. If c then d. If d then e. This creates a chain of logic.':

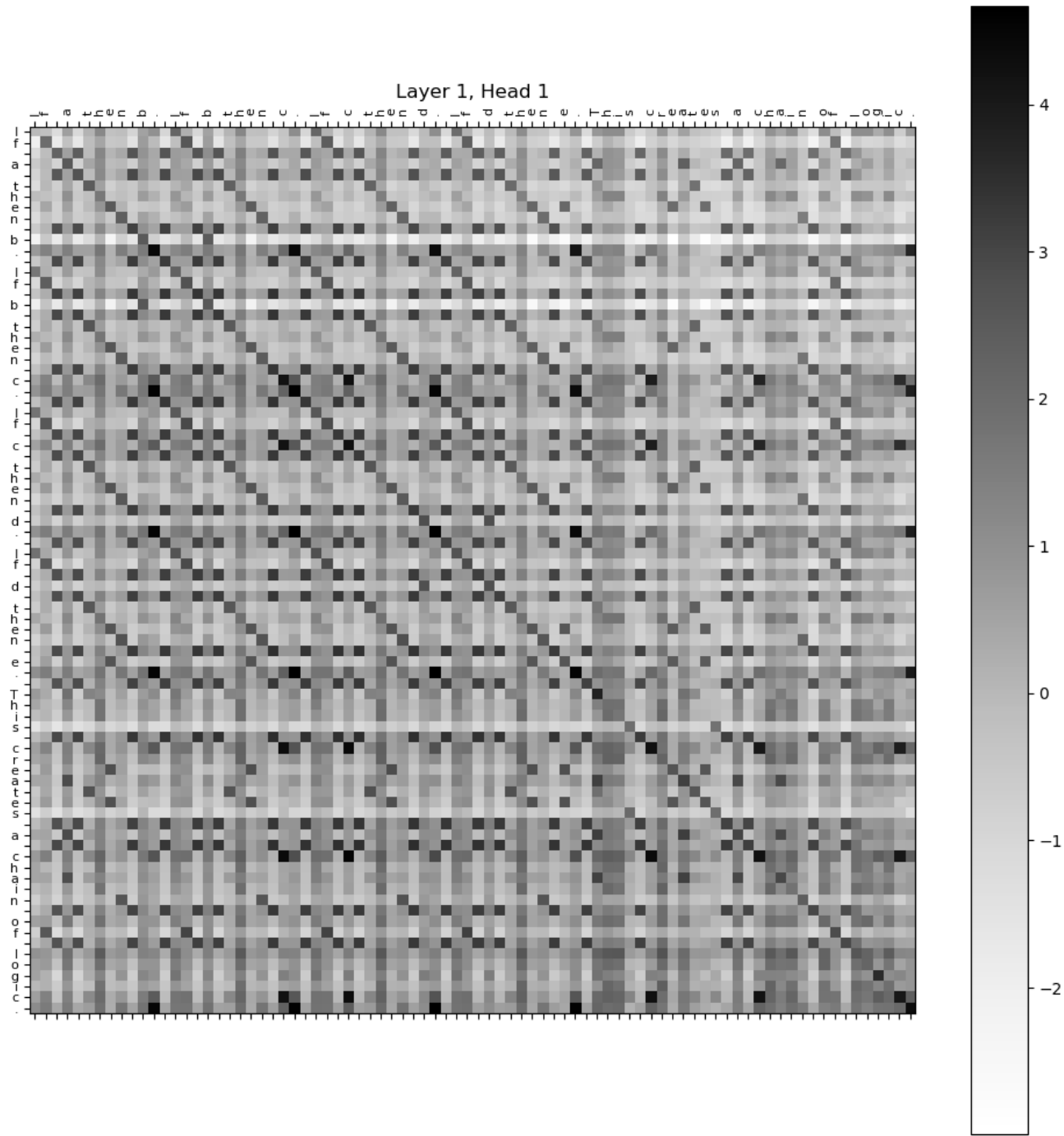
For examples:

- Head 1: Focuses on **logical connectors**, strongly attending between `if` → `then`
- Head 4: Focuses on the beginning of words.
- Head 5: Focuses on the whitespaces.

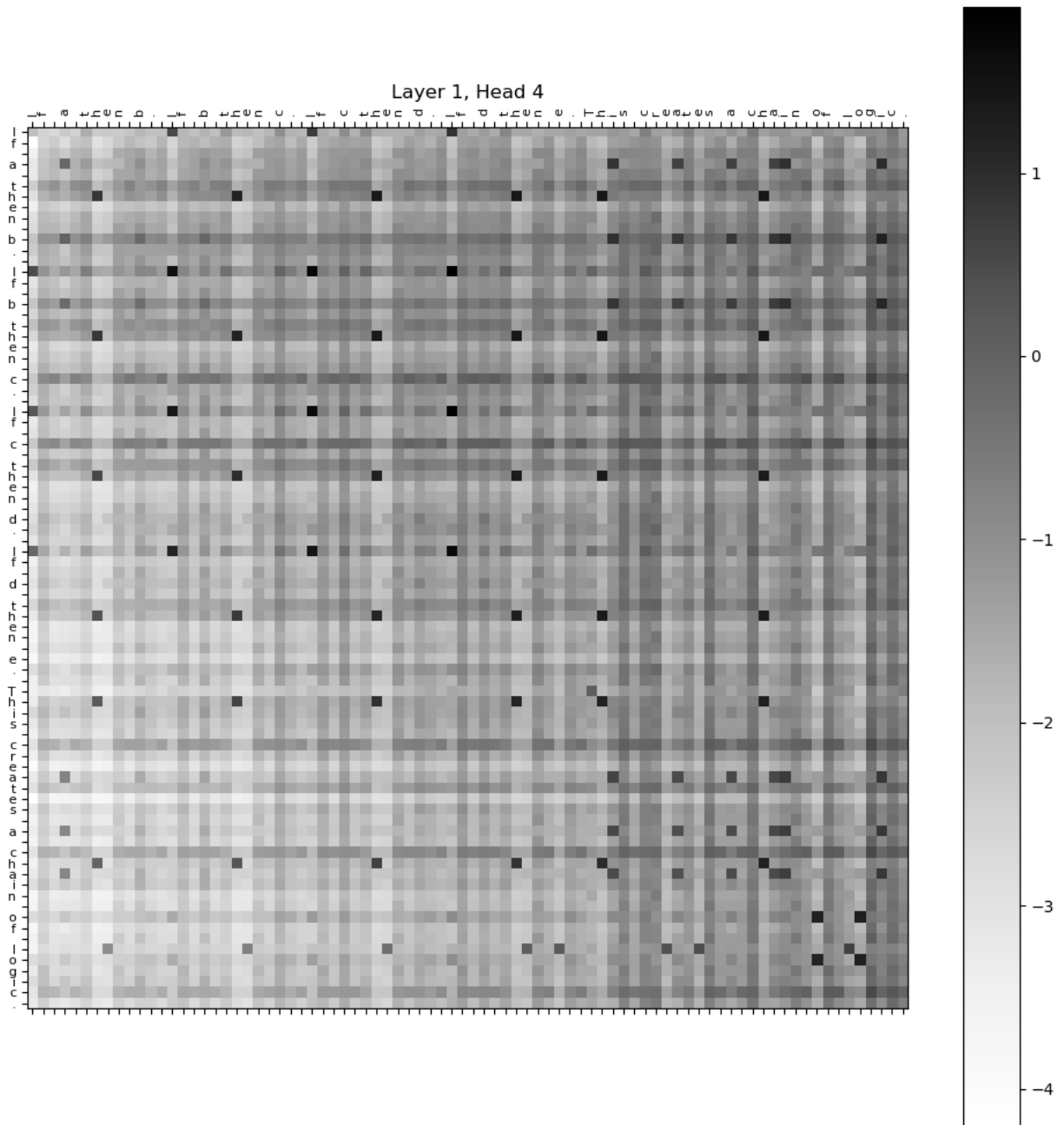
As you can see in the following figures.

Attention Heatmaps for Layer 1 Heads

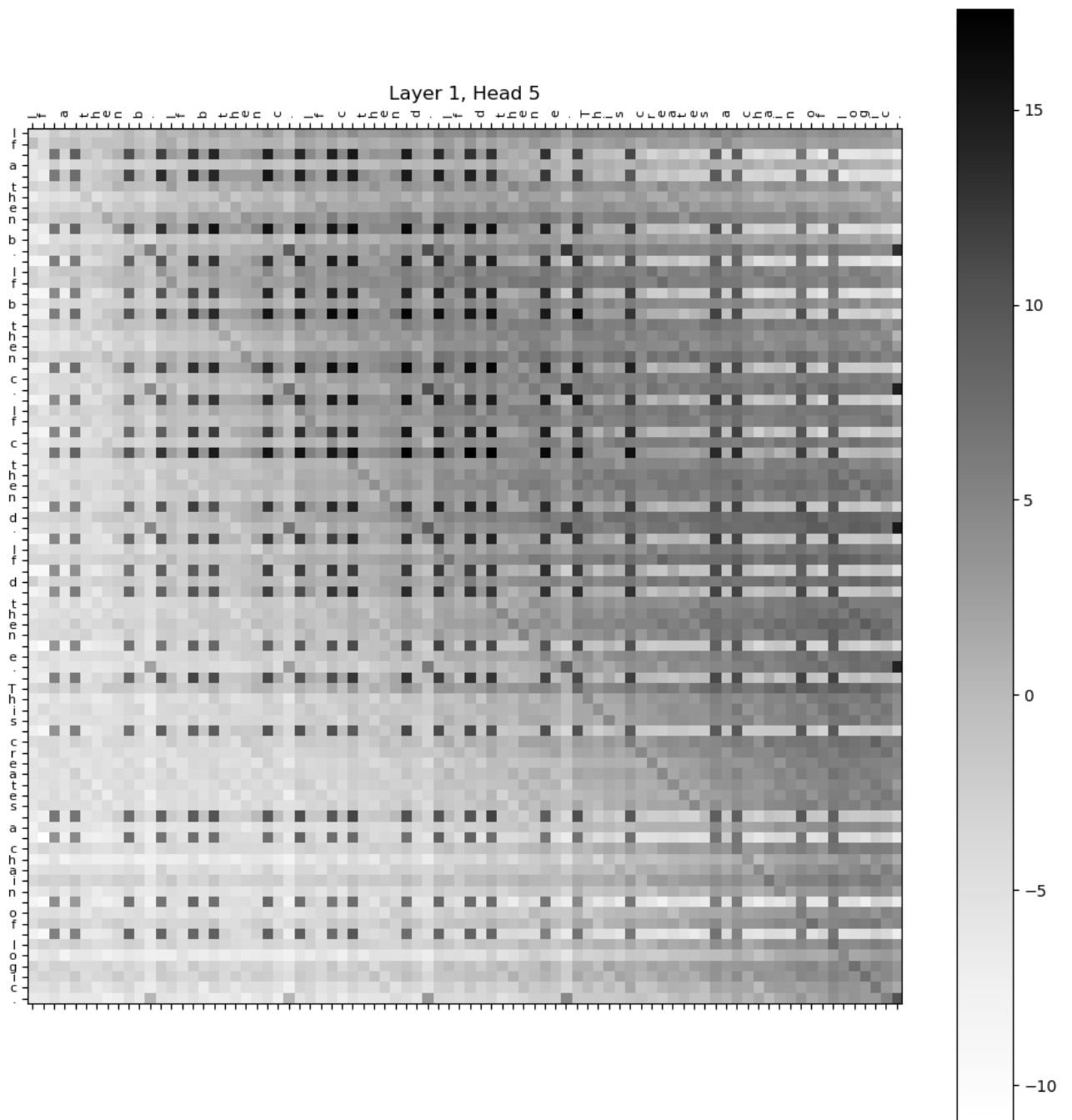
- **Head 1**: Focuses on logical relationships.



- **Head 4**: Focuses on the beginning of words.



- **Head 5**: Focuses on the whitespaces



6. Conclusion

This project allowed us to develop and train a Transformer language model at the character level, focusing on understanding its attention mechanisms. Analyzing attention patterns helped us gain insights into how the model processes linguistic structures, improving our skills in model development and interpretability.